MAKERERE UNIVERSITY | COLLEGE OF COMPUTING & INFORMATION SCIENCES

COURSEWORK DELIVERABLES & DEFENSE PREPARATION GUIDE

Report Structure, Wokwi Simulation Setup, Video Scripts & Viva Q&A

Kayeera Nathan (24/U/25913/EVE) - Group Leader (Group SE-24)

COURSEWORK DELIVERABLES & DEFENSE PREPARATION GUIDE

Student: Kayeera Nathan (Group Leader & System Architect, SE-24)

Project: Smart Control and Monitoring System for CoCIS, Makerere University

Communication Bus: Wireless ESP-NOW (2.4 GHz Peer-to-Peer Wi-Fi)

Target Deadline: 2 November 2026

1. Overview of Required Deliverables (Per Coursework Brief)

According to Section 4 of the official coursework specification, your group must submit a comprehensive package consisting of:

- 1. System Operation Report (PDF on MUELE):
- Architecture, hardware pinout, software architecture, FreeRTOS synchronization, wireless ESP-NOW protocol, and feature documentation.
- Individual contributions breakdown.
- Group member details.
- 2. Wokwi Simulation Link: Live, publicly accessible simulation demonstrating the multi-board wireless system.
- 3. Individual Video Demonstrations: Hyperlinks to 5 individual screen recordings (3-5 minutes each).
- 4. Viva / Oral Presentation: Monday, 2 November 2026 before the lecturer (attendance compulsory for marks).

2. Deliverable 1: Report Structure & Author Assignments

To ensure maximum marks under the assessment criteria, structure the final report as follows:

```
Title Page: Project Title, Course, Academic Year, Date, Group SE-24 Details Table
1. Executive Summary & Problem Background ( Nathan )
2. System Architecture & Distributed Wireless Topology ( Nathan )
   - 3-ESP32 Distributed Architecture
   - Peer-to-Peer 2.4 GHz ESP-NOW Wireless Mesh/Star
   - Hardware Pin Mapping & Circuit Schematics
3. Communication & Synchronization Protocols ( Nathan )
   - ESP-NOW Connectionless Packet Framing & Delivery ACKs
   - FreeRTOS Mutexes, Semaphores & Message Queues
   - Race Condition & Concurrency Prevention
4. Entrance Terminal & Access Control Subsystem ( David )
   - RFID Reading & Door Mechanism (Servos)
   - Schedule Enforcement & Anti-Passback Algorithm
5. Lecture Room Automated Lighting Subsystem ( Mark )
   - PIR Motion Sensing, Debounce & Timers
   - Ambient Light Thresholding (LDR) & ESP-NOW Telemetry Dispatch
6. Data Storage & System Reliability Subsystem ( Jovan )
   - SPI SD Card Implementation & FATFS Filesystem
   - CSV Schemas & Crash Recovery from Power Failure
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7. Serial Command Interface & Security Management ( Laurinda )
   - Command Parser & Navigation Menus
   - Authentication (Hashed Passwords, Session Timeout, Lockout)
8. Innovative Features & System Value-Add ( Nathan & All )
   - Industrial Wireless IoT Network (ESP-NOW) over Cable-based Serial (Nathan)
   - Wireless Emergency Evacuation Broadcast (Nathan)
   - Wireless Heartbeat & Node Failure Detection (Nathan)
   - Anti-Passback Security (David)
   - Daylight-Aware Energy Saving Logic (Mark)
   - Crash Recovery & Non-Volatile State Restoration (Jovan)
   - Serial Intrusion Lockout (Laurinda)
9. Verification, Test Results & Concurrency Evaluation ( Laurinda & All )
   - Comprehensive Test Matrix (Inputs, Expected vs. Observed Outputs)
10. System Links, Contributions & Group Member Declarations
   - Public Wokwi Project Link
   - Individual Video Links Table
   - Itemized Individual Contributions Table
   - Declaration of Originality
```

3. Deliverable 2: Wokwi Multi-ESP32 Wireless Simulation Setup

In Wokwi, you can simulate multiple microcontrollers communicating via ESP-NOW in a single project. Because ESP-NOW operates over the simulated 2.4 GHz radio space, no physical RX/TX interconnecting wires are required between the boards!

Clean Wokwi ‘diagram.json‘ Setup:

```
{
  "version": 1,
  "author": "Kayeera Nathan",
  "editor": "wokwi",
  "parts": [
    { "type": "board-esp32-devkit-c-v4", "id": "enterance_master_esp", "top": 0, "left": 0, "at
    { "type": "board-esp32-devkit-c-v4", "id": "room_node_1_esp",      "top": 0, "left": 420, "
    { "type": "board-esp32-devkit-c-v4", "id": "room_node_2_esp",      "top": 0, "left": 840, "
  ],
  "connections": []
}
```

TIP / BEST PRACTICE

Why Wokwi Multi-ESP32 Wireless is Elegant: Each ESP32 board initializes Wi-Fi in Station mode and uses ‘esp_now_send()‘ with the target’s MAC address. Wokwi’s virtual RF environment routes packets directly between the simulated ESP32 instances.

4. Deliverable 3: Nathan’s Individual Video Demonstration Script

Every student must submit a 3 to 5 minute video. Marks are awarded for clarity and demonstration of code you wrote.

Recommended Video Outline (4 Minutes Total):

- 0:00 - 0:45 (Introduction & Architecture Overview):
- "Hello, I am Kayeera Nathan, registration number 24/U/25913/EVE. I am the group leader, system architect, and integrator for group SE-24. Our project is the Smart Control and Monitoring System for CoCIS."
- "Our architecture is distributed across three ESP32 nodes communicating wirelessly over ESP-NOW, Espressif's connectionless 2.4 GHz Wi-Fi protocol, eliminating the need for hundreds of meters of fragile serial cabling across university corridors."

- 0:45 - 2:00 (Architecture, Synchronization & FreeRTOS Walkthrough):
- Display common/protocol.h and sync_manager.c.
- Explain: "Under Section 2.7, resource synchronization is paramount. I designed the FreeRTOS concurrency model where incoming wireless packets are pushed into xWirelessPacketQueue directly from the ESP-NOW receive callback ISR, preventing task starvation. Shared state-including our building occupancy table and operating schedule-is guarded by mutexes (xOccupancyMutex, xScheduleMutex) to eliminate race conditions."
- 2:00 - 3:00 (Wireless ESP-NOW Protocol & Node Heartbeat):
- Show the C code in esp_now_transport.c.
- Explain the 2.4 GHz packet framing, MAC addressing, and hardware delivery status callbacks.
- Demonstrate the master's periodic heartbeat task that monitors Room 1 and Room 2 telemetry every 2 seconds and flags an alert if a wireless node drops offline.
- 3:00 - 4:15 (Live Wokwi Demonstration of Wireless Emergency Evacuation):
- Switch to the live Wokwi browser tab.
- Run the simulation; type status or rooms in the master terminal to show live wireless telemetry streaming from the lecture rooms.
- Trigger emergency on in the terminal.
- Point out: "Observe that both entrance door servos instantly swing to 90 degrees open. Simultaneously, an emergency broadcast packet is transmitted over ESP-NOW, causing both lecture room microcontrollers to override their lights ON for occupant evacuation, and an emergency event is logged to the SD card."
- 4:15 - 4:30 (Conclusion):
- Summarize the reliability and innovations of the system.

5. Defense & Presentation Q&A Preparation

The lecturer will question each member individually on Monday, 2 November. As Group Leader and Architect, expect questions like:

- 1. "Why did you choose ESP-NOW over traditional UART serial cables?"
- *Answer:* "In a campus building like CoCIS, routing physical RS232/UART serial cables from the main entrance to multiple lecture rooms across floors is expensive, susceptible to electromagnetic interference, and difficult to scale. ESP-NOW provides a low-latency, cable-free 2.4 GHz wireless solution with built-in hardware MAC-layer delivery confirmation and zero reliance on external network infrastructure."
- 2. "Why did you use ESP-NOW instead of standard Wi-Fi with MQTT or HTTP?"
- *Answer:* "Standard Wi-Fi requires connecting to a router (SSID/password), dynamic IP allocation via DHCP, and an external MQTT broker or cloud server. If campus Wi-Fi or internet goes down, building access and safety would be compromised. ESP-NOW is completely connectionless and operates peer-to-peer at the 802.11 MAC layer, guaranteeing sub-5ms latency and full offline autonomy."
- 3. "How does your system prevent race conditions when a wireless packet arrives during an SD card write?"
- *Answer:* "The ESP-NOW receive callback executes in an interrupt context. Instead of performing heavy operations inside the callback, it copies the packet into a FreeRTOS queue (xWirelessPacketQueue) using xQueueSendFromISR(). A separate worker task consumes the queue and acquires xRoomStateMutex before updating shared memory. Meanwhile, the SD card logger runs in its own task guarded by xSDCardMutex, completely decoupling storage I/O from wireless reception."
- 4. "How does your system enforce that people inside cannot leave outside hours?"
- *Answer:* "The exit door logic checks both the schedule and the user's role. If a student attempts to scan out after curfew (e.g. 20:00), the state machine evaluates current_time > schedule_end. Unless the tag has administrative credentials, the exit servo remains locked and the event is flagged in access_log.csv as an unauthorized after-hours attempt."